

ARM-450 rev I

Assembly manual -- 2026-09-24

6-DOF arm, 6 x Waveshare ST3215 on one bus.
27 printed files (33 pieces), 199 bought items, 114 screws.
Verified: 26 release stages at 0 failures (LOG_SUMMARY.pdf).

READ FIRST

- 1 Print the fit test (7 coupons). Coupon 07 must screw onto the back of your servo before the 6 servo housings are printed.
- 2 Fix the foot to the table (4 holes) -- it tips at reach.
- 3 Every servo is screwed by its OWN back holes with the self-tapping screws from its box (2.6 mm go in; a longer screw needs a washer).
- 4 Plug both bus cables in before closing a servo cover; leave the loop given for each joint (last page).
- 5 Bearings are held by press fit: a drop of retaining compound on each outer ring.

The build order below is the one a screwdriver was proven to follow: every screw reachable at its step.
BENCH = a sub-assembly on the table; JOIN = they go together.

1 The six joints

joint	parent	child	range
J1_base_ya	ground	turret	+ -90.0 deg
J2_shoulde	turret	upper_link	+ -54.0 deg
J3_elbow	upper_link	forearm	+ -72.0 deg
J4_forearm	forearm	j4_hub	+ -90.0 deg
J5_wrist_p	j4_hub	blade	+ -46.5 deg
J6_tool_ro	blade	flange	+ -90.0 deg

Axis points / directions (mm, world at home): EDITABLE_CAD/JOINT_AXES.csv.
Set these ranges as the servo limits in your firmware; home (straight up)
is singular -- start from a bent ready pose (J2 30, J3 -60, J5 30).

2 Bought hardware (counted from the placed assembly)

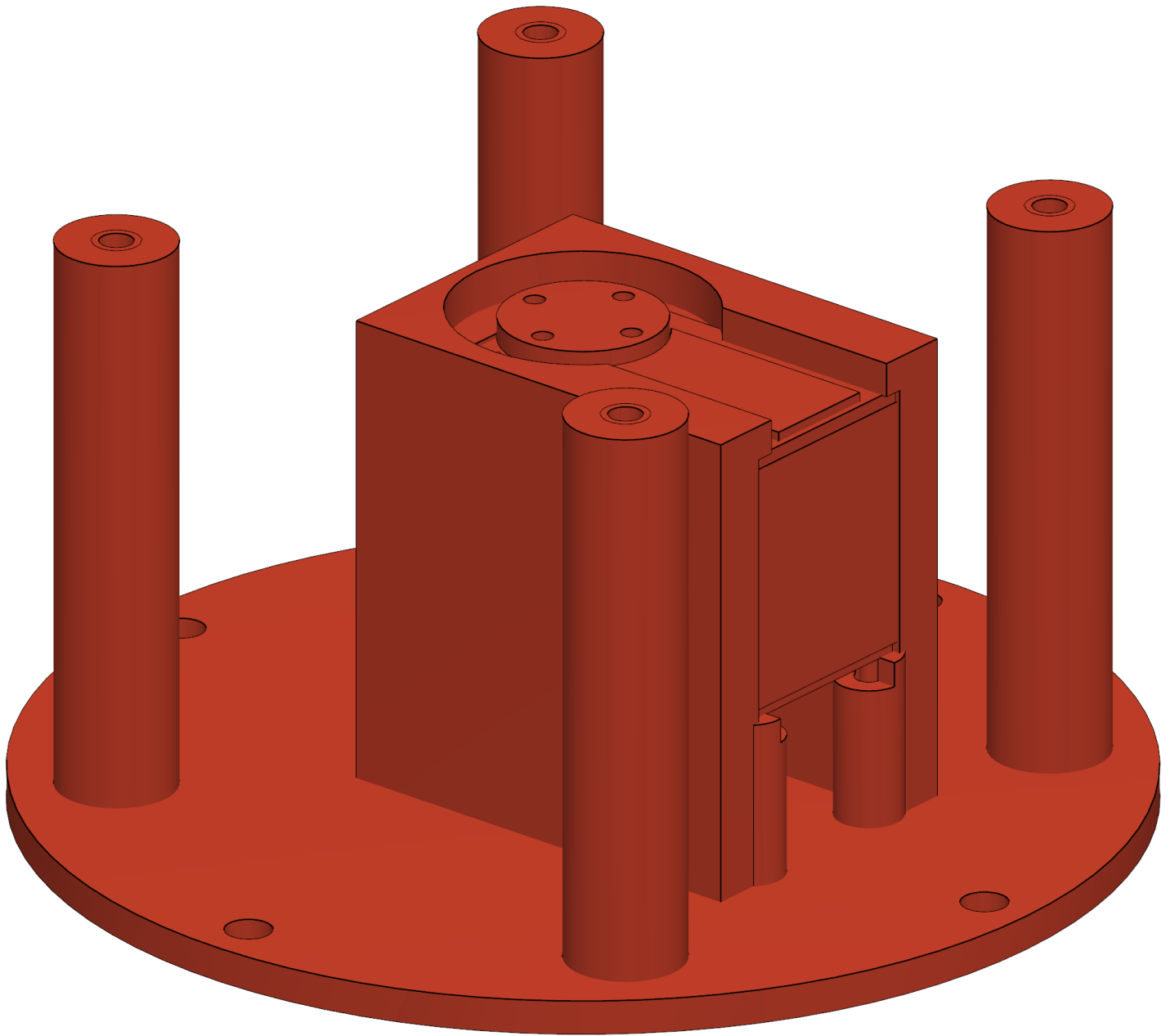
item	qty
6706 bearing (30x37x4)	4
6806 bearing (30x42x7)	6
M2.5 heat-set insert	2
M2.5 x 16 socket head	2
M2.5 x 6 socket head (horn)	24
M3 heat-set insert	55
M3 nyloc nut	4
M3 x 10 socket head	12
M3 x 12 socket head	16
M3 x 16 socket head	10
M3 x 20 socket head	1
M3 x 4 set screw	1
M3 x 45 socket head	16
M4 heat-set insert	4
M4 x 12 socket head	4
M5 x 10 set screw (cup point)	8
ST3215 servo	6
extension spring J2 (7.4 N @ 36.0 / 12.4 N @ 58.2 mm, insi	2
extension spring J3 (2.2 N @ 43.0 / 5.0 N @ 91.4 mm, insid	2
self-tapping screw ~2.2 x 8 pan head (from the servo box)	20

3 Print list

file	qty	orient	supports	size mm
01_base_x1.stl	1	+Z up	REQUIRED	120 x 120 x 50
02_spigot_collar_x1.stl	1	+Z up	none	46 x 46 x 12
03_j1_mount_x1.stl	1	+Z up	none	128 x 128 x 70
04_j1_hub_x1.stl	1	+Z up	advised	29 x 29 x 17
05_J2_turret_p1_x1.stl	1	+X up	REQUIRED	138 x 70 x 84
06_J2_turret_p2_x1.stl	1	-Y up	advised	83 x 66 x 37
07_J2_shaft_x1.stl	1	+Y up	none	30 x 30 x 50
08_shaft_clamp_x4.stl	4	-Z up	none	37 x 38 x 14
09_link_upper_groove_x1.stl	1	+Z up	advised	110 x 50 x 14
10_link_upper_tongue_x1.stl	1	+Z up	advised	110 x 50 x 16
11_collar_upper_x2.stl	2	+Z up	none	64 x 24 x 10
12_J3_p1_x1.stl	1	-Z up	REQUIRED	98 x 50 x 53
13_J3_p2_x1.stl	1	-Z up	none	72 x 50 x 37
14_J3_shaft_x1.stl	1	-Z up	none	30 x 30 x 50
15_link_fore_tongue_x1.stl	1	+Z up	advised	144 x 50 x 16
16_link_fore_groove_x1.stl	1	+Z up	advised	144 x 50 x 14
17_collar_fore_x2.stl	2	+Z up	none	64 x 24 x 10
18_j4_base_x1.stl	1	-Z up	REQUIRED	57 x 38 x 65
19_j4_cap_x1.stl	1	-Z up	none	58 x 45 x 10
20_j4_hub_x1.stl	1	-Z up	none	46 x 46 x 20
21_J5_p1_x1.stl	1	+X up	advised	48 x 50 x 66
22_J5_p2_x1.stl	1	-Z up	none	66 x 50 x 37
23_J5_shaft_x1.stl	1	-Z up	none	30 x 30 x 45
24_J5_spacer_x2.stl	2	+Z up	none	32 x 32 x 3
25_j6_body_x1.stl	1	-X up	REQUIRED	43 x 54 x 87
26_j6_cap_x1.stl	1	-X up	none	45 x 58 x 10
27_j6_flange_x1.stl	1	-X up	none	46 x 46 x 20

00_FIT_TEST_print_first: 7 coupons, print FIRST (see PRINTABLE_FILES).

Step 1 of 24 -- foot: J1 servo into the foot, its 4 screws from under the foot plate

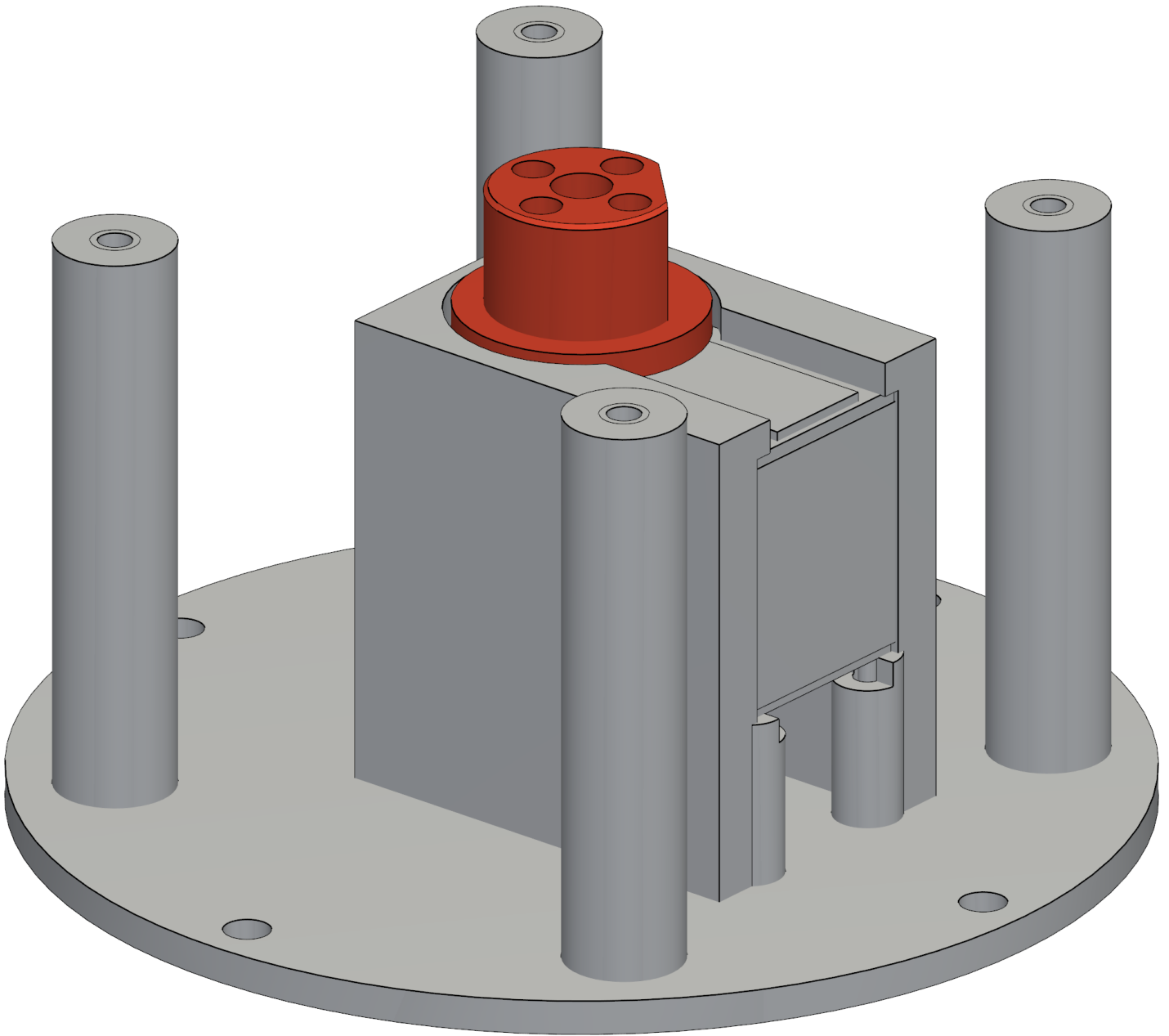


Before this step: drive 4 x M5 (or #10 wood) screws through the foot plate's 05.5 holes into the table -- the arm tips over at reach otherwise. Set the 4 M4 inserts in the column tops. Slide the J1 servo into the channel from the open end until it stops on the end wall (its front face under the lips), then drive its 4 self-tapping screws from UNDER the foot plate into the servo's back holes.

ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M4 heat-set insert
- 1 x ST3215 servo
- printed: j1_mount
- 4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

Step 2 of 24 -- J1 hub onto the horn



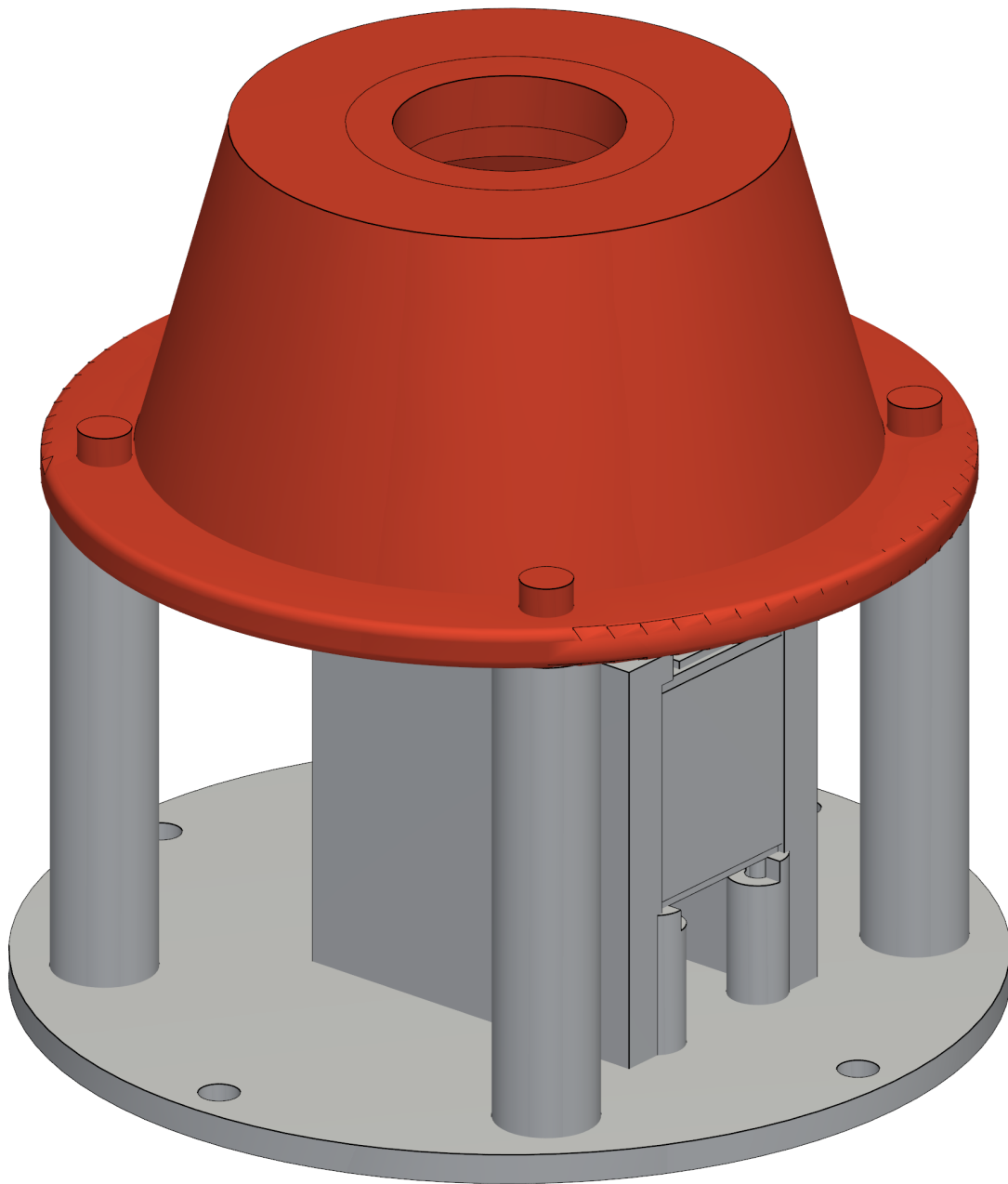
4 x M2.5 x 6 through the hub flange into the horn.

ADDED IN THIS STEP (red in the picture, grey = already built):

4 x M2.5 x 6 socket head (horn)

printed: j1_hub

Step 3 of 24 -- base + J1 bearings onto the foot



Press the two 6806 into the base; base onto the 4 columns, 4 x M4 x 12.

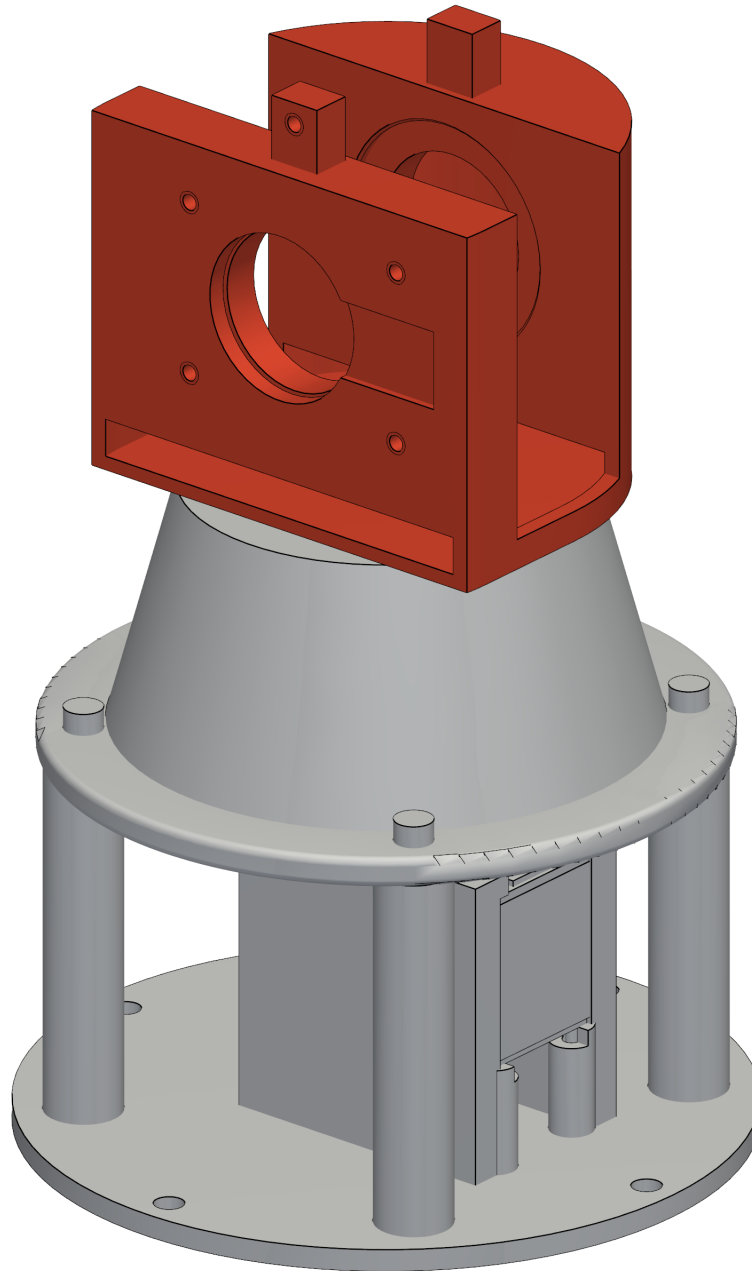
ADDED IN THIS STEP (red in the picture, grey = already built):

- 2 x 6806 bearing (30x42x7)

- 4 x M4 x 12 socket head

- printed: base

Step 4 of 24 -- turret p1 + J2 bearings, spigot collar



Press the 6806 pair into turret p1 from the fork gap; set the inserts. Turret spigot down through the base bearings onto the hub plug (the D-flat keys it). Spigot collar round the slit spigot end; tighten its pinch bolt LAST (it closes the collet onto the plug).

ADDED IN THIS STEP (red in the picture, grey = already built):

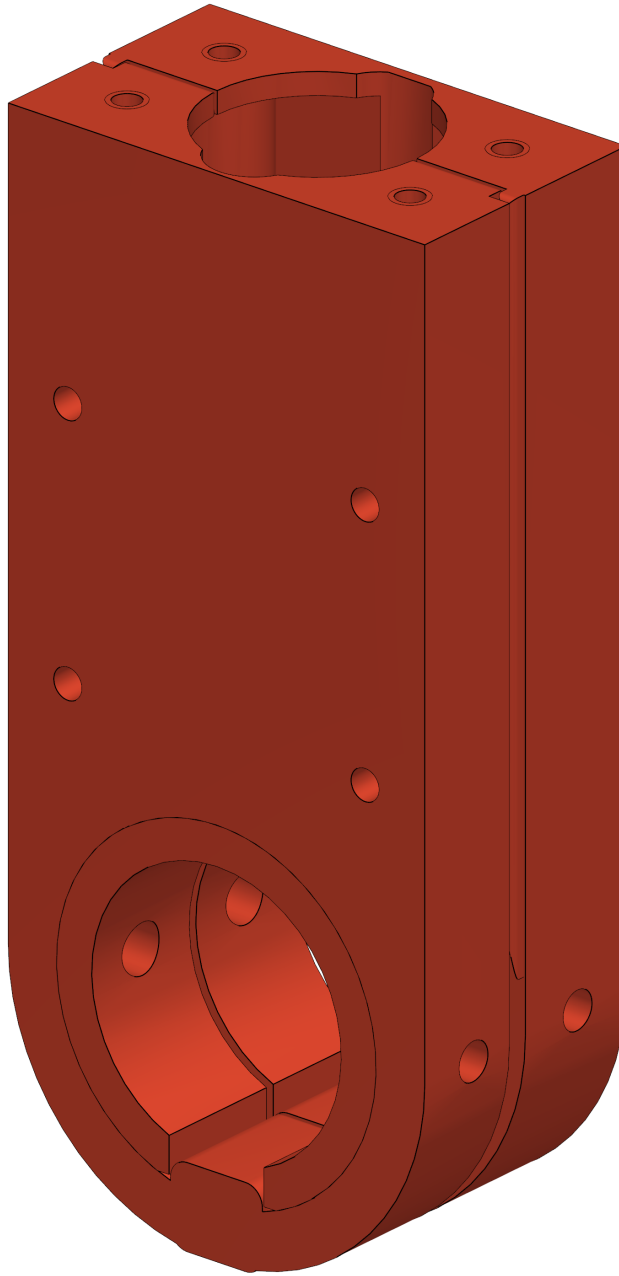
- 2 x 6806 bearing (30x42x7)

- 7 x M3 heat-set insert

- 1 x M3 x 20 socket head

printed: J2_turret_p1, spigot_collar

Step 5 of 24 -- BENCH upper link



Upper link: centring rings into both halves (gap over the seam ear), run an M5 into each Ø4.2 side hole once to form the thread, then close the halves with the seam screws (heads in the tongue's counterbores).

ADDED IN THIS STEP (red in the picture, grey = already built):

- 1 x M2.5 heat-set insert

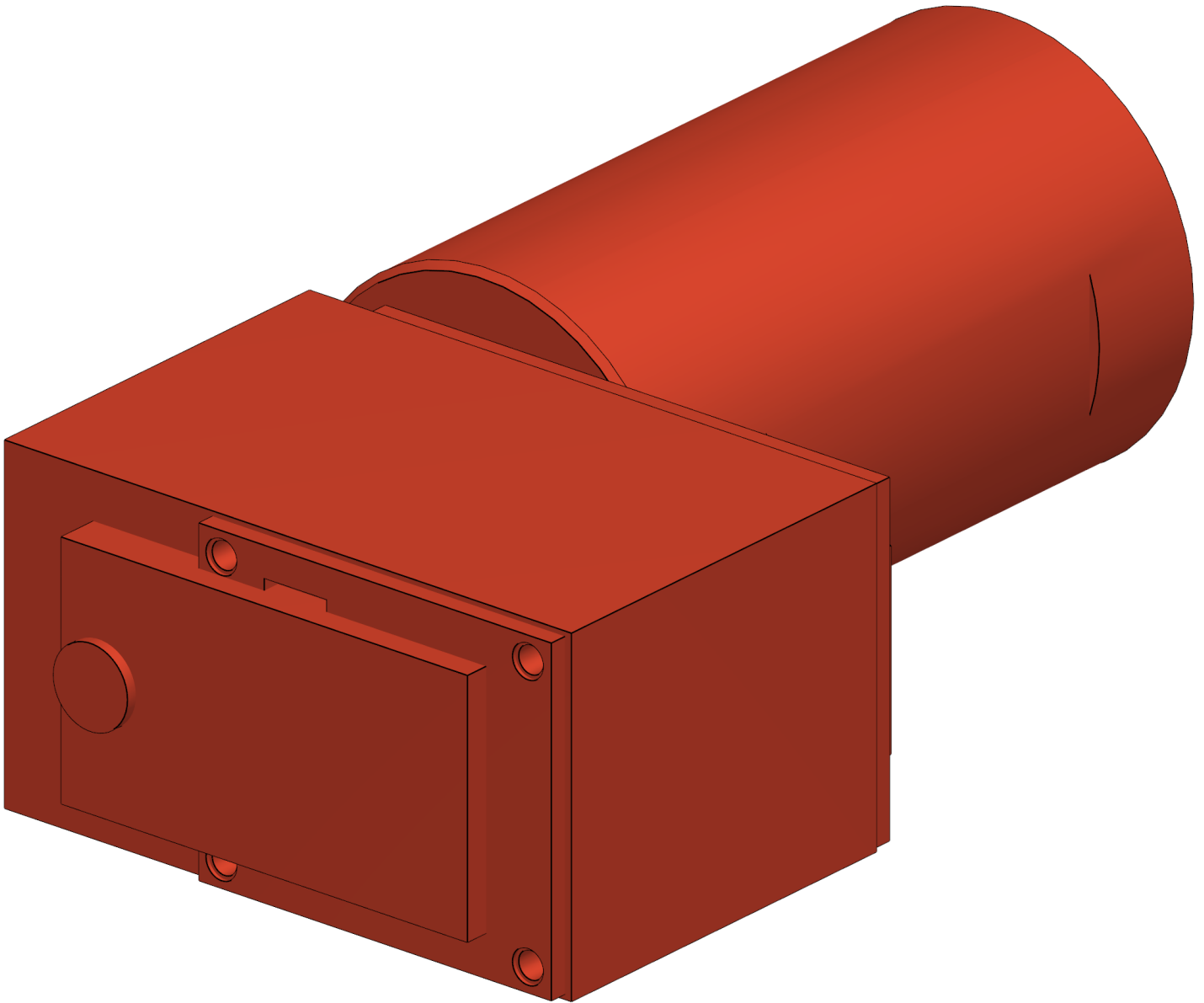
- 1 x M2.5 x 16 socket head

- 8 x M3 heat-set insert

- 4 x M3 x 16 socket head

printed: link_upper_groove, link_upper_tongue, shaft_clamp_1, shaft_clamp_2

Step 6 of 24 -- BENCH J2 servo + shaft



J2 servo onto the printed shaft: 4 x M2.5 through the shaft bore into the horn.

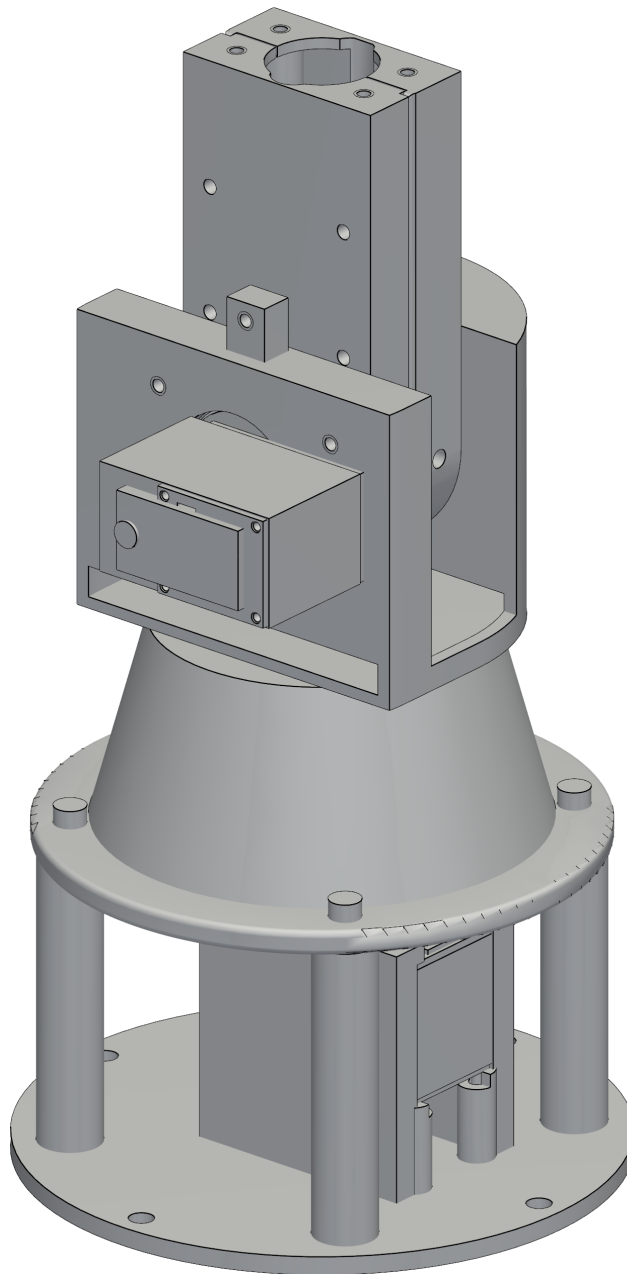
ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M2.5 x 6 socket head (horn)

- 1 x ST3215 servo

- printed: J2_shaft

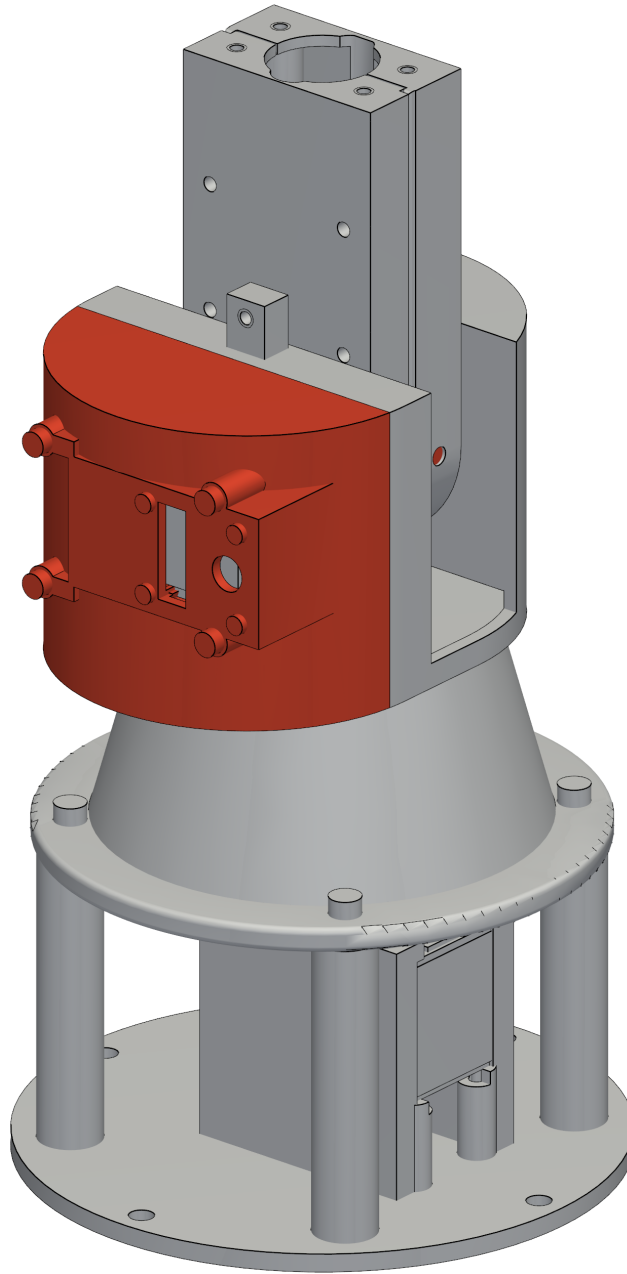
Step 7 of 24 -- JOIN link into the J2 gap, servo+shaft down the axis



Put the link into the J2 gap and the servo + shaft down the axis through the link's rings and the bearings.

ADDED IN THIS STEP (red in the picture, grey = already built):

Step 8 of 24 -- J2 pod on, servo screws, link set screws



Close the turret pod: 4 x M3 x 45. Then 4 self-tapping servo screws through the pod's back into the servo. Lock the link: 2 x M5 x 10 cup-point set screws per half onto the shaft's flats.

ADDED IN THIS STEP (red in the picture, grey = already built):

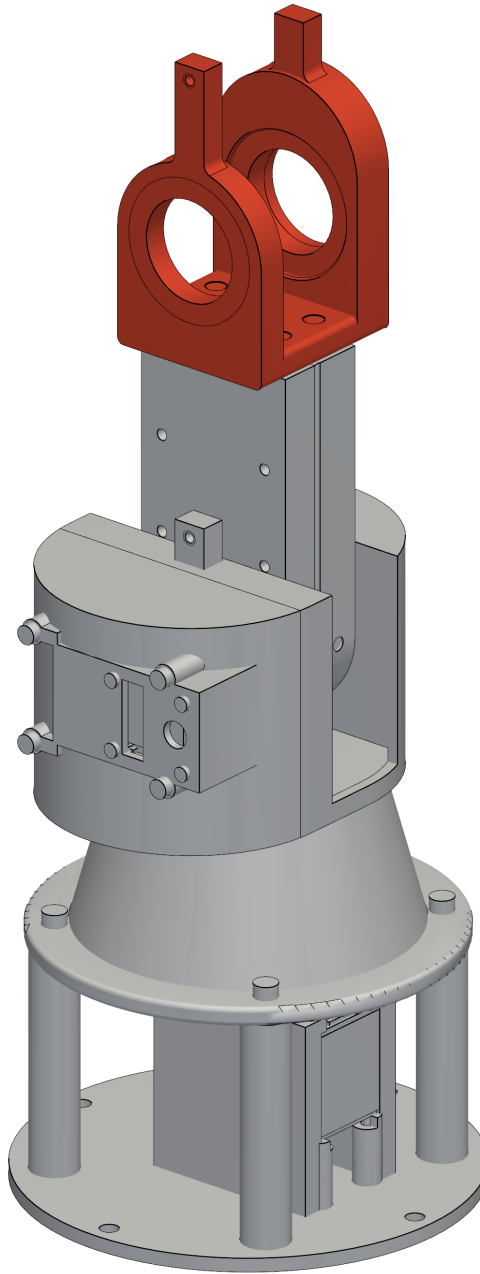
- 4 x M3 x 45 socket head

- 4 x M5 x 10 set screw (cup point)

- printed: J2_turret_p2

- 4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

Step 9 of 24 -- J3 fork p1 onto the upper link



J3 fork p1 onto the upper link's socket end: 4 x M3 x 10, heads sunk in the fork floor.

ADDED IN THIS STEP (red in the picture, grey = already built):

- 2 x 6806 bearing (30x42x7)

- 6 x M3 heat-set insert

- 4 x M3 x 10 socket head

printed: J3_p1

Step 10 of 24 -- BENCH forearm



Forearm halves round their centring rings, as the upper link.

ADDED IN THIS STEP (red in the picture, grey = already built):

- 1 x M2.5 heat-set insert

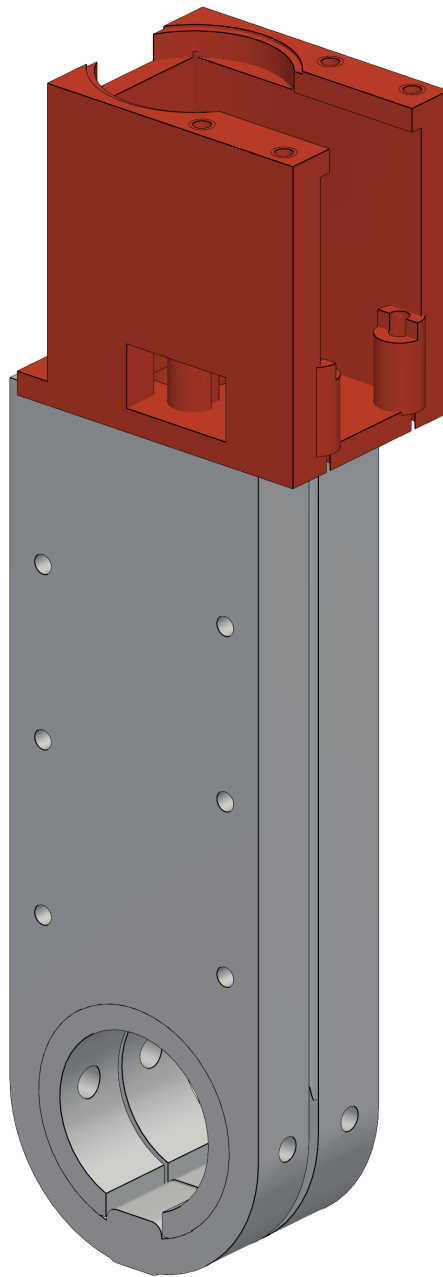
- 1 x M2.5 x 16 socket head

- 10 x M3 heat-set insert

- 6 x M3 x 16 socket head

printed: link_fore_groove, link_fore_tongue, shaft_clamp_3, shaft_clamp_4

Step 11 of 24 -- BENCH J4 base on the forearm

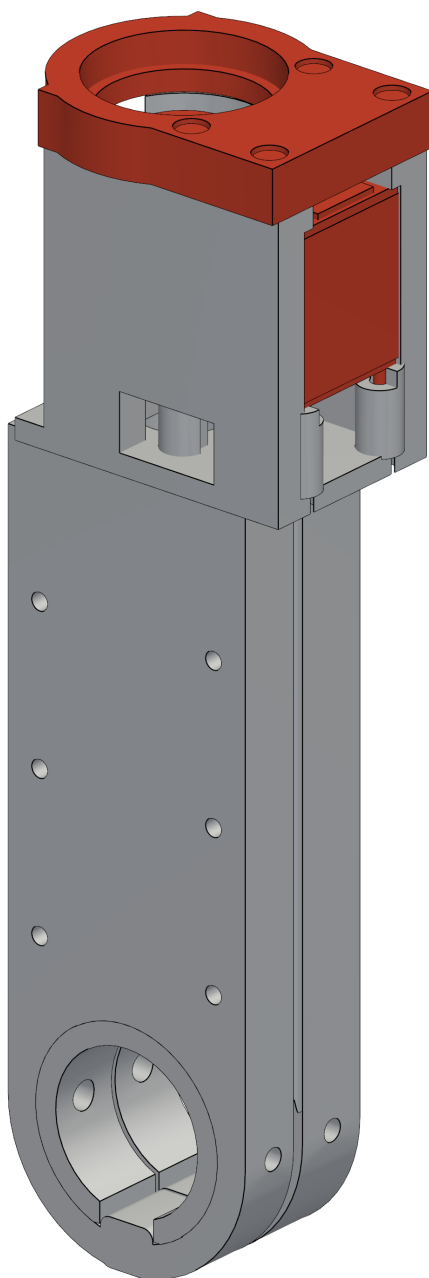


J4 base onto the forearm's socket end: 4 x M3 x 10 (heads on the plate).

ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M3 heat-set insert
- 4 x M3 x 10 socket head
- printed: j4_base

Step 12 of 24 -- BENCH J4 servo (2 screws from under the plate), bearing, cap



J4 servo into its channel; 2 self-tapping screws from UNDER the base plate into its far back holes. 6706 into the cap; cap onto the walls: 4 x M3 x 12 (heads in the 3.8 mm counterbores).

ADDED IN THIS STEP (red in the picture, grey = already built):

- 1 x 6706 bearing (30x37x4)

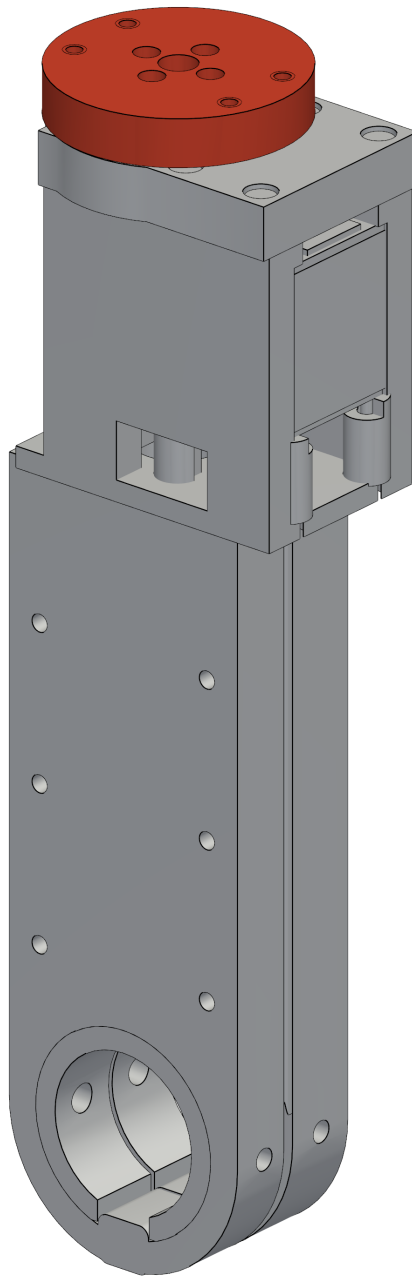
- 4 x M3 x 12 socket head

- 1 x ST3215 servo

- printed: j4_cap

- 2 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

Step 13 of 24 -- BENCH J4 hub onto the horn



J4 hub onto the horn: 4 x M2.5.

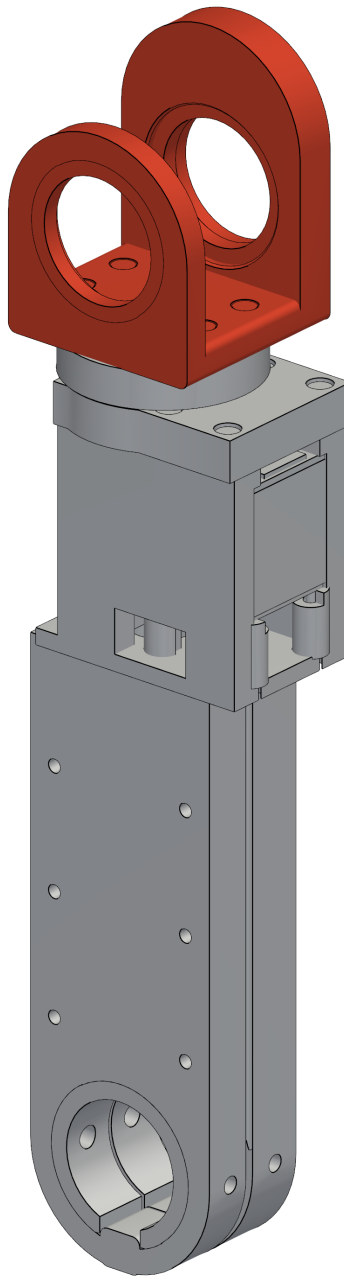
ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M2.5 x 6 socket head (horn)

- 4 x M3 heat-set insert

- printed: j4_hub

Step 14 of 24 -- BENCH J5 fork p1 onto the hub



J5 fork p1 onto the J4 hub: 4 x M3 x 10, heads sunk in the fork floor.

ADDED IN THIS STEP (red in the picture, grey = already built):

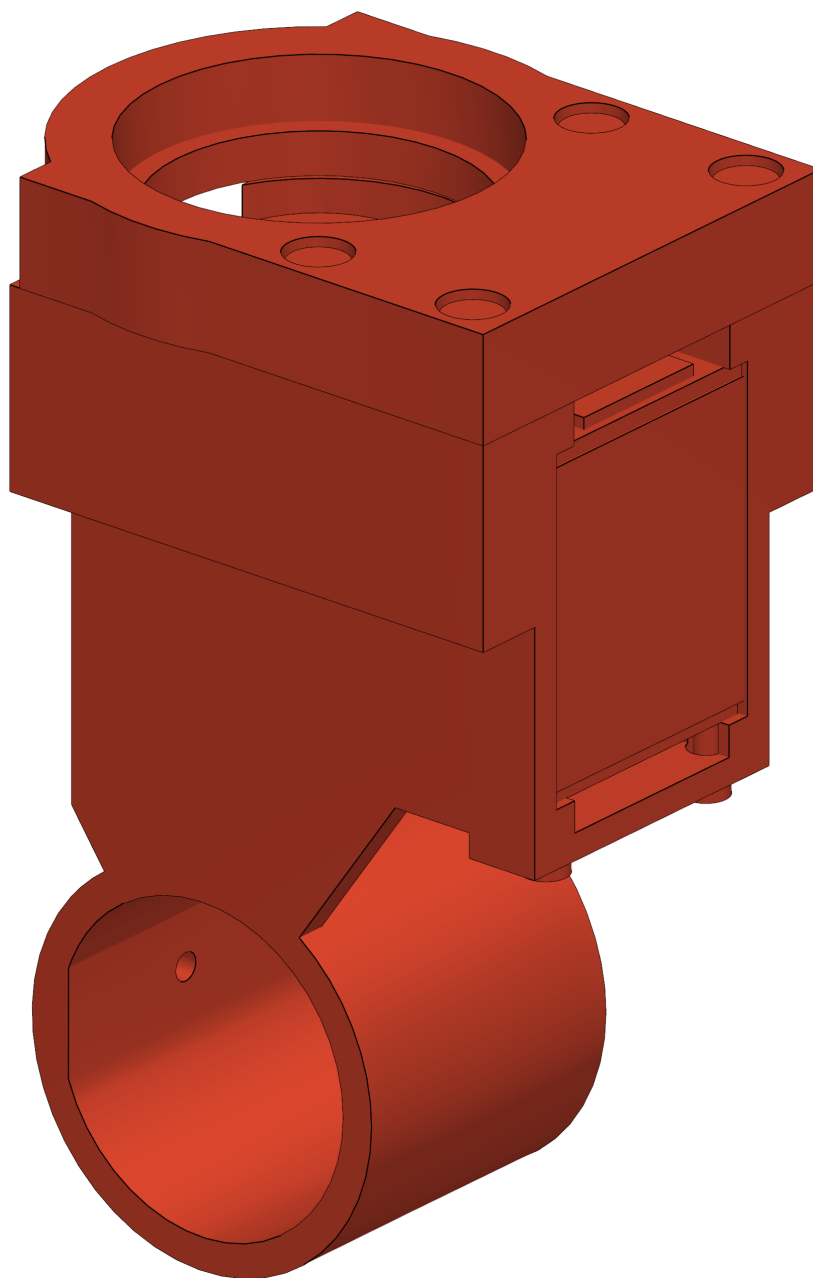
- 2 x 6706 bearing (30x37x4)

- 4 x M3 heat-set insert

- 4 x M3 x 10 socket head

printed: J5_p1

Step 15 of 24 -- BENCH blade: J6 servo, bearing, cap

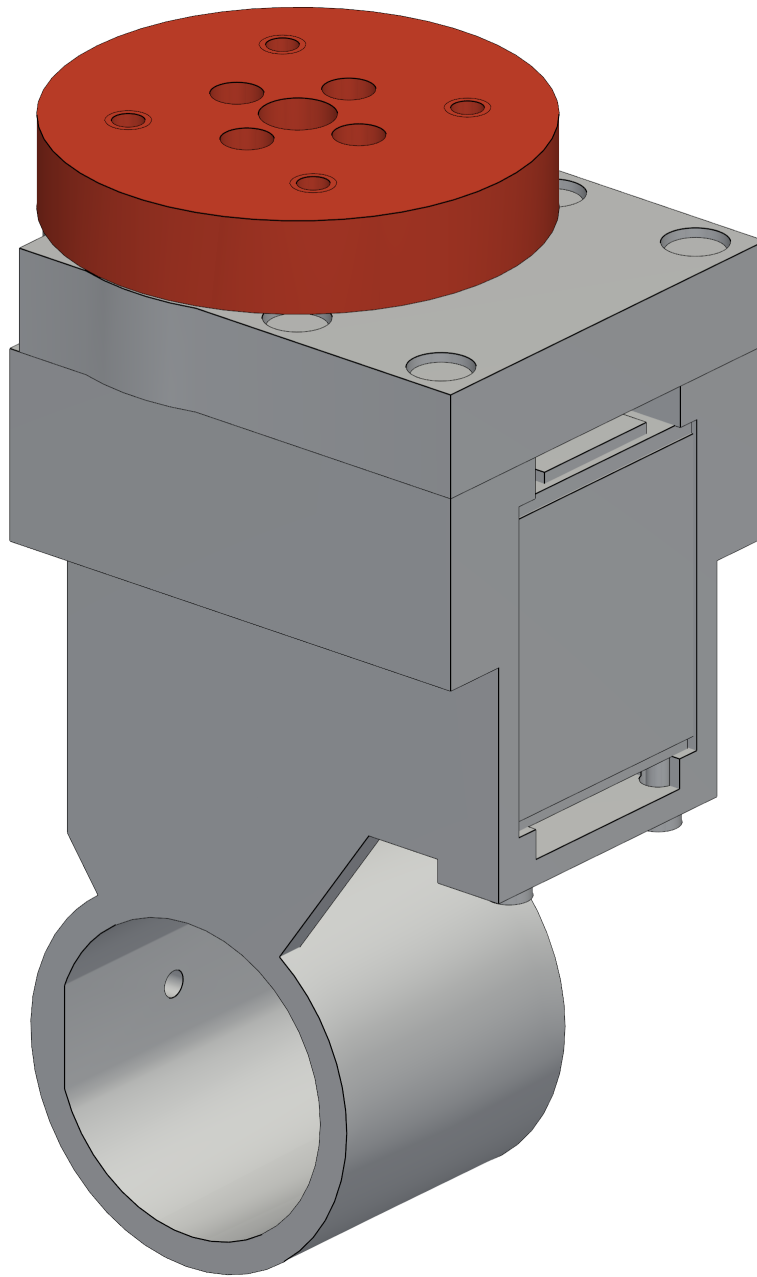


J6 servo into the blade's channel; 2 self-tapping screws through the bridge into its far back holes. 6706 into the cap; cap: 4 x M3 x 12.

ADDED IN THIS STEP (red in the picture, grey = already built):

- 1 x 6706 bearing (30x37x4)
- 4 x M3 heat-set insert
- 4 x M3 x 12 socket head
- 1 x ST3215 servo
- printed: j6_body, j6_cap
- 2 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

Step 16 of 24 -- BENCH blade: tool flange onto the J6 horn



Tool flange onto the J6 horn: 4 x M2.5.

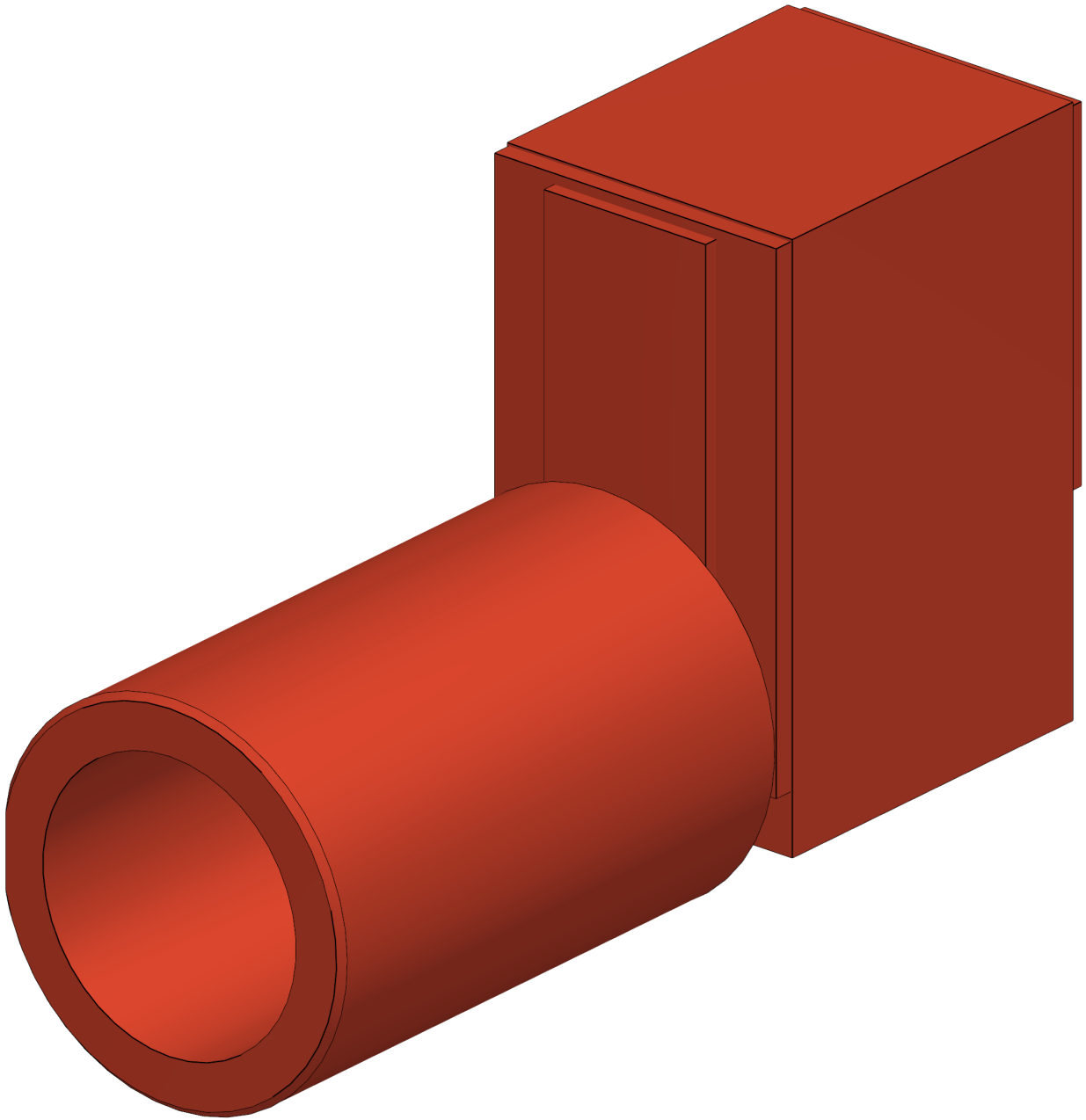
ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M2.5 x 6 socket head (horn)

- 4 x M3 heat-set insert

- printed: j6_flange

Step 17 of 24 -- BENCH J5 servo + axle



J5 servo onto its printed axle: 4 x M2.5.

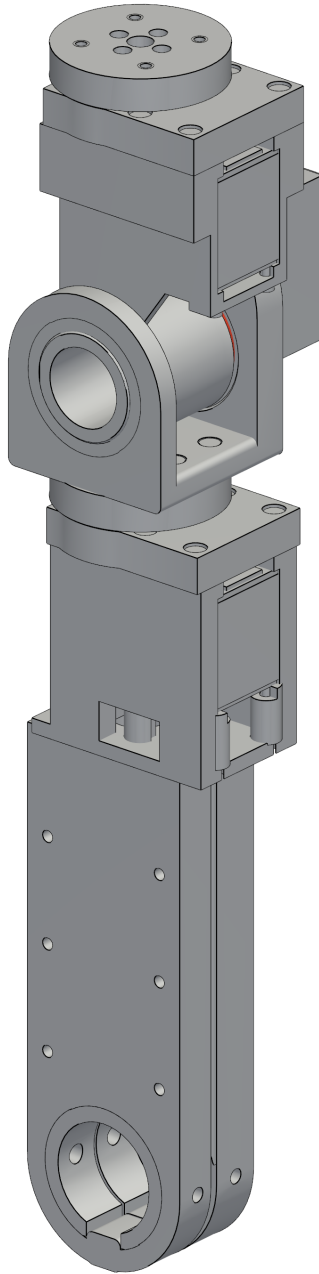
ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M2.5 x 6 socket head (horn)

- 1 x ST3215 servo

- printed: J5_shaft

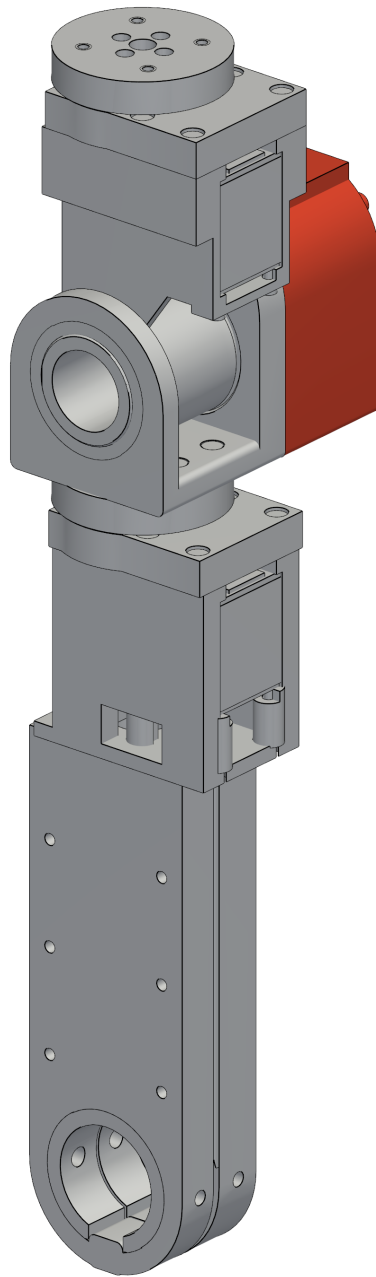
Step 18 of 24 -- JOIN blade into the J5 gap, spacers, axle down the axis



Blade into the J5 gap with the two 3.0 mm spacers; the axle goes in from the idle side, its flat to the blade's D-bore.

ADDED IN THIS STEP (red in the picture, grey = already built):
printed: J5_spacer_A, J5_spacer_B

Step 19 of 24 -- J5 cover, servo screws, axle grub

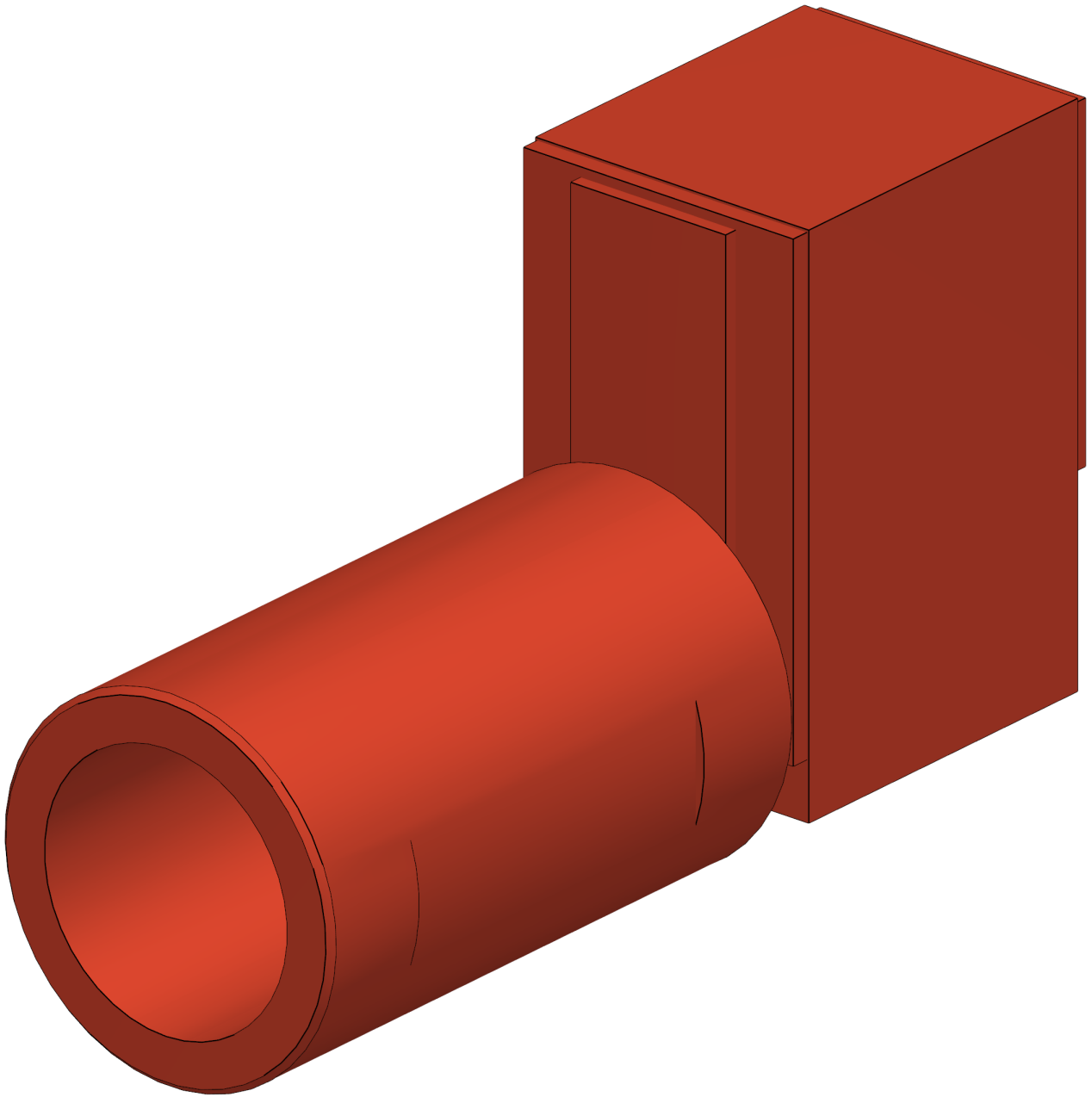


Close the J5 cover: 4 x M3 x 45, 4 servo screws through the cover; M3 x 4 grub on the axle's flat.

ADDED IN THIS STEP (red in the picture, grey = already built):

- 1 x M3 x 4 set screw
- 4 x M3 x 45 socket head
- printed: J5_p2
- 4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

Step 20 of 24 -- BENCH J3 servo + shaft



J3 servo onto its shaft: 4 x M2.5.

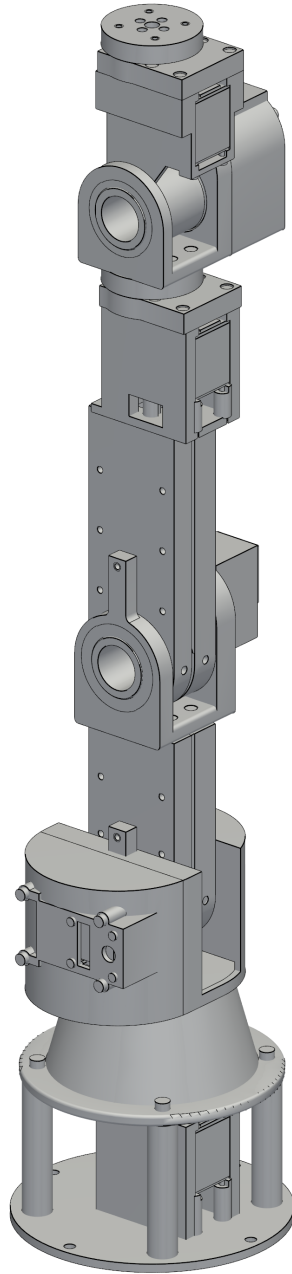
ADDED IN THIS STEP (red in the picture, grey = already built):

- 4 x M2.5 x 6 socket head (horn)

- 1 x ST3215 servo

- printed: J3_shaft

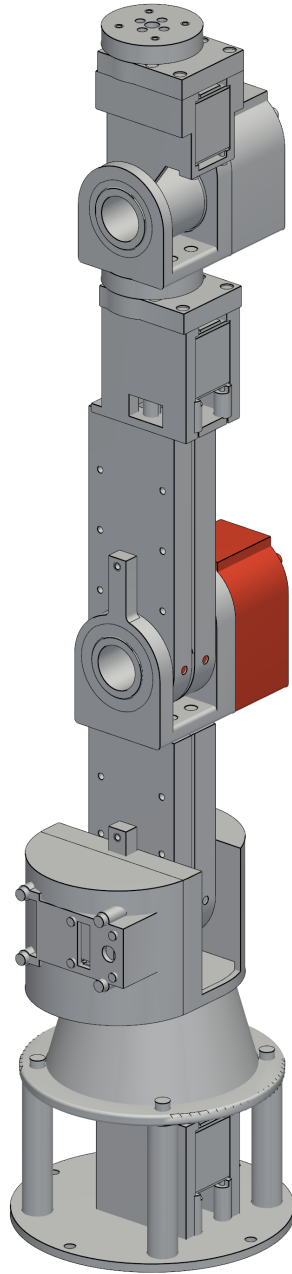
Step 21 of 24 -- JOIN forearm into the J3 gap, servo+shaft down the axis



Forearm into the J3 gap, servo + shaft down the axis.

ADDED IN THIS STEP (red in the picture, grey = already built):

Step 22 of 24 -- J3 cover, servo screws, link set screws



Close the J3 cover: 4 x M3 x 45, 4 servo screws; lock the forearm: 2 x M5 x 10 set screws per half.

ADDED IN THIS STEP (red in the picture, grey = already built):

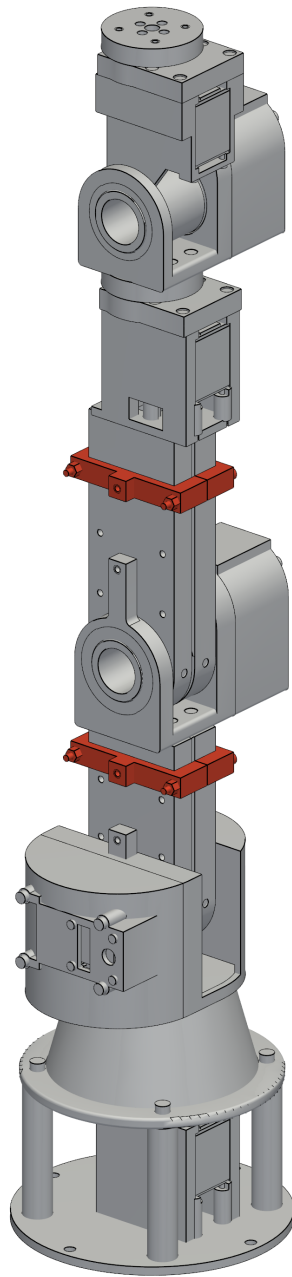
- 4 x M3 x 45 socket head

- 4 x M5 x 10 set screw (cup point)

- printed: J3_p2

- 4 x self-tapping screw ~2.2 x 8 pan head (from the servo box)

Step 23 of 24 -- spring collars



Spring collars on the upper link and the forearm: M3 x 45 + nyloc.

ADDED IN THIS STEP (red in the picture, grey = already built):

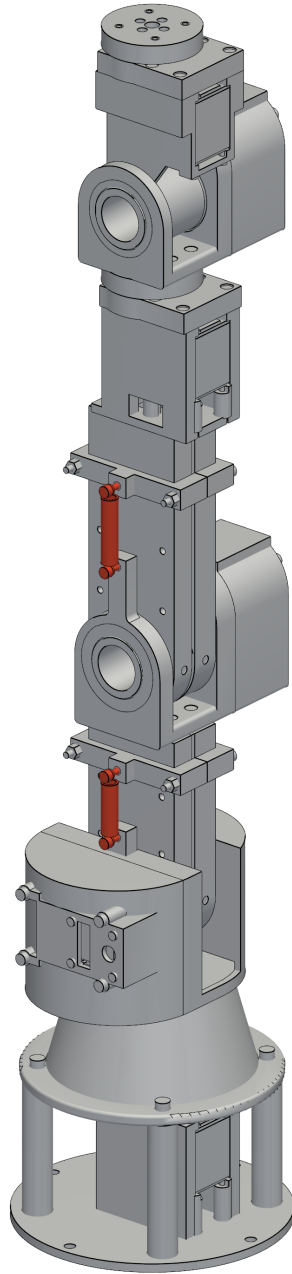
- 4 x M3 heat-set insert

- 4 x M3 nyloc nut

- 4 x M3 x 45 socket head

printed: collar_fore_+y, collar_fore_-y, collar_upper_+y, collar_upper_-y

Step 24 of 24 -- spring pins + springs



Spring pins (M3 x 12 shoulder screws) and the 4 gravity springs.

ADDED IN THIS STEP (red in the picture, grey = already built):

- 8 x M3 x 12 socket head

- 2 x extension spring J2 (7.4 N @ 36.0 / 12.4 N @ 58.2 mm, inside hooks)

- 2 x extension spring J3 (2.2 N @ 43.0 / 5.0 N @ 91.4 mm, inside hooks)

4 Fastenerschedule

screw	size	step	head on	holds in / access
servo_scr_J1_0	self-tapping screw ~2.2 x 8 pan head (fr	1	j1_mount	servo_J1 / clear
servo_scr_J1_1	self-tapping screw ~2.2 x 8 pan head (fr	1	j1_mount	servo_J1 / clear
servo_scr_J1_2	self-tapping screw ~2.2 x 8 pan head (fr	1	j1_mount	servo_J1 / clear
servo_scr_J1_3	self-tapping screw ~2.2 x 8 pan head (fr	1	j1_mount	servo_J1 / clear
horn_screw_J1_0	M2.5 x 6 socket head (horn)	2	j1_hub	j1_hub,servo_J1 / clear
horn_screw_J1_180	M2.5 x 6 socket head (horn)	2	j1_hub	j1_hub,servo_J1 / clear
horn_screw_J1_270	M2.5 x 6 socket head (horn)	2	j1_hub	j1_hub,servo_J1 / clear
horn_screw_J1_90	M2.5 x 6 socket head (horn)	2	j1_hub	j1_hub,servo_J1 / clear
m4_base_foot_0	M4 x 12 socket head	3	base	j1_mount / clear
m4_base_foot_1	M4 x 12 socket head	3	base	j1_mount / clear
m4_base_foot_2	M4 x 12 socket head	3	base	j1_mount / covered later
m4_base_foot_3	M4 x 12 socket head	3	base	j1_mount / covered later
collar_pinch	M3 x 20 socket head	4	spigot_collar	spigot_collar / clear
ear_bolt_upper	M2.5 x 16 socket head	5	link_upper_tongu	link_upper_groov / covered later
seam_upper_0	M3 x 16 socket head	5	link_upper_tongu	link_upper_groov / covered later
seam_upper_1	M3 x 16 socket head	5	link_upper_tongu	link_upper_groov / covered later
seam_upper_2	M3 x 16 socket head	5	link_upper_tongu	link_upper_groov / clear
seam_upper_3	M3 x 16 socket head	5	link_upper_tongu	link_upper_groov / clear
horn_screw_J2_0	M2.5 x 6 socket head (horn)	6	J2_shaft	J2_shaft,servo_J / covered later
horn_screw_J2_180	M2.5 x 6 socket head (horn)	6	J2_shaft	J2_shaft,servo_J / covered later
horn_screw_J2_270	M2.5 x 6 socket head (horn)	6	J2_shaft	J2_shaft,servo_J / covered later
horn_screw_J2_90	M2.5 x 6 socket head (horn)	6	J2_shaft	J2_shaft,servo_J / covered later
link_set_1_270	M5 x 10 set screw (cup point)	8	-	- / clear
link_set_1_90	M5 x 10 set screw (cup point)	8	-	- / clear
link_set_2_270	M5 x 10 set screw (cup point)	8	-	- / clear
link_set_2_90	M5 x 10 set screw (cup point)	8	-	- / clear
servo_scr_J2_0	self-tapping screw ~2.2 x 8 pan head (fr	8	J2_turret_p2	servo_J2 / clear
servo_scr_J2_1	self-tapping screw ~2.2 x 8 pan head (fr	8	J2_turret_p2	servo_J2 / clear
servo_scr_J2_2	self-tapping screw ~2.2 x 8 pan head (fr	8	J2_turret_p2	servo_J2 / clear
servo_scr_J2_3	self-tapping screw ~2.2 x 8 pan head (fr	8	J2_turret_p2	servo_J2 / clear
split_J2_0	M3 x 45 socket head	8	J2_turret_p2	J2_turret_p1 / clear
split_J2_1	M3 x 45 socket head	8	J2_turret_p2	J2_turret_p1 / clear
split_J2_2	M3 x 45 socket head	8	J2_turret_p2	J2_turret_p1 / clear
split_J2_3	M3 x 45 socket head	8	J2_turret_p2	J2_turret_p1 / clear
j3fork_link_0	M3 x 10 socket head	9	J3_p1	link_upper_groov / covered later
j3fork_link_1	M3 x 10 socket head	9	J3_p1	link_upper_tongu / covered later
j3fork_link_2	M3 x 10 socket head	9	J3_p1	link_upper_groov / covered later
j3fork_link_3	M3 x 10 socket head	9	J3_p1	link_upper_tongu / covered later
ear_bolt_fore	M2.5 x 16 socket head	10	link_fore_tongue	link_fore_groove / covered later
seam_fore_0	M3 x 16 socket head	10	link_fore_tongue	link_fore_groove / covered later
seam_fore_1	M3 x 16 socket head	10	link_fore_tongue	link_fore_groove / covered later
seam_fore_2	M3 x 16 socket head	10	link_fore_tongue	link_fore_groove / clear
seam_fore_3	M3 x 16 socket head	10	link_fore_tongue	link_fore_groove / clear
seam_fore_4	M3 x 16 socket head	10	link_fore_tongue	link_fore_groove / covered later
seam_fore_5	M3 x 16 socket head	10	link_fore_tongue	link_fore_groove / covered later
j4base_forearm_0	M3 x 10 socket head	11	j4_base	link_fore_groove / covered later
j4base_forearm_1	M3 x 10 socket head	11	j4_base	link_fore_tongue / covered later
j4base_forearm_2	M3 x 10 socket head	11	j4_base	link_fore_groove / covered later
j4base_forearm_3	M3 x 10 socket head	11	j4_base	link_fore_tongue / covered later
cap_J4_0	M3 x 12 socket head	12	j4_cap	j4_base / covered later
cap_J4_1	M3 x 12 socket head	12	j4_cap	j4_base / covered later
cap_J4_2	M3 x 12 socket head	12	j4_cap	j4_base / clear
cap_J4_3	M3 x 12 socket head	12	j4_cap	j4_base / clear
servo_scr_J4_0	self-tapping screw ~2.2 x 8 pan head (fr	12	j4_base	servo_J4 / covered later
servo_scr_J4_1	self-tapping screw ~2.2 x 8 pan head (fr	12	j4_base	servo_J4 / covered later
horn_screw_J4_0	M2.5 x 6 socket head (horn)	13	j4_hub	j4_hub,servo_J4 / covered later
horn_screw_J4_180	M2.5 x 6 socket head (horn)	13	j4_hub	j4_hub,servo_J4 / covered later
horn_screw_J4_270	M2.5 x 6 socket head (horn)	13	j4_hub	j4_hub,servo_J4 / covered later
horn_screw_J4_90	M2.5 x 6 socket head (horn)	13	j4_hub	j4_hub,servo_J4 / covered later
j5fork_j4hub_0	M3 x 10 socket head	14	J5_p1	j4_hub / covered later
j5fork_j4hub_1	M3 x 10 socket head	14	J5_p1	j4_hub / covered later
j5fork_j4hub_2	M3 x 10 socket head	14	J5_p1	j4_hub / covered later
j5fork_j4hub_3	M3 x 10 socket head	14	J5_p1	j4_hub / covered later
cap_J6_0	M3 x 12 socket head	15	j6_cap	j6_body / covered later
cap_J6_1	M3 x 12 socket head	15	j6_cap	j6_body / covered later
cap_J6_2	M3 x 12 socket head	15	j6_cap	j6_body / clear
cap_J6_3	M3 x 12 socket head	15	j6_cap	j6_body / clear
servo_scr_J6_0	self-tapping screw ~2.2 x 8 pan head (fr	15	j6_body	servo_J6 / clear
servo_scr_J6_1	self-tapping screw ~2.2 x 8 pan head (fr	15	j6_body	servo_J6 / clear
horn_screw_J6_0	M2.5 x 6 socket head (horn)	16	j6_flange	j6_flange,servo_ / clear

4 Fastener schedule (cont.)

screw	size	step	head on	holds in / access
horn_screw_J6_180	M2.5 x 6 socket head (horn)	16	j6_flange	j6_flange,servo_ / clear
horn_screw_J6_270	M2.5 x 6 socket head (horn)	16	j6_flange	j6_flange,servo_ / clear
horn_screw_J6_90	M2.5 x 6 socket head (horn)	16	j6_flange	j6_flange,servo_ / clear
horn_screw_J5_0	M2.5 x 6 socket head (horn)	17	J5_shaft	J5_shaft,servo_J / clear
horn_screw_J5_180	M2.5 x 6 socket head (horn)	17	J5_shaft	J5_shaft,servo_J / clear
horn_screw_J5_270	M2.5 x 6 socket head (horn)	17	J5_shaft	J5_shaft,servo_J / clear
horn_screw_J5_90	M2.5 x 6 socket head (horn)	17	J5_shaft	J5_shaft,servo_J / clear
j5_grub	M3 x 4 set screw	19	-	- / clear
servo_scr_J5_0	self-tapping screw ~2.2 x 8 pan head (fr	19	J5_p2	servo_J5 / clear
servo_scr_J5_1	self-tapping screw ~2.2 x 8 pan head (fr	19	J5_p2	servo_J5 / clear
servo_scr_J5_2	self-tapping screw ~2.2 x 8 pan head (fr	19	J5_p2	servo_J5 / clear
servo_scr_J5_3	self-tapping screw ~2.2 x 8 pan head (fr	19	J5_p2	servo_J5 / clear
split_J5_0	M3 x 45 socket head	19	J5_p2	J5_p1 / clear
split_J5_1	M3 x 45 socket head	19	J5_p2	J5_p1 / clear
split_J5_2	M3 x 45 socket head	19	J5_p2	J5_p1 / clear
split_J5_3	M3 x 45 socket head	19	J5_p2	J5_p1 / clear
horn_screw_J3_0	M2.5 x 6 socket head (horn)	20	J3_shaft	J3_shaft,servo_J / clear
horn_screw_J3_180	M2.5 x 6 socket head (horn)	20	J3_shaft	J3_shaft,servo_J / clear
horn_screw_J3_270	M2.5 x 6 socket head (horn)	20	J3_shaft	J3_shaft,servo_J / clear
horn_screw_J3_90	M2.5 x 6 socket head (horn)	20	J3_shaft	J3_shaft,servo_J / clear
link_set_3_270	M5 x 10 set screw (cup point)	22	-	- / clear
link_set_3_90	M5 x 10 set screw (cup point)	22	-	- / clear
link_set_4_270	M5 x 10 set screw (cup point)	22	-	- / clear
link_set_4_90	M5 x 10 set screw (cup point)	22	-	- / clear
servo_scr_J3_0	self-tapping screw ~2.2 x 8 pan head (fr	22	J3_p2	servo_J3 / clear
servo_scr_J3_1	self-tapping screw ~2.2 x 8 pan head (fr	22	J3_p2	servo_J3 / clear
servo_scr_J3_2	self-tapping screw ~2.2 x 8 pan head (fr	22	J3_p2	servo_J3 / clear
servo_scr_J3_3	self-tapping screw ~2.2 x 8 pan head (fr	22	J3_p2	servo_J3 / clear
split_J3_0	M3 x 45 socket head	22	J3_p2	J3_p1 / clear
split_J3_1	M3 x 45 socket head	22	J3_p2	J3_p1 / clear
split_J3_2	M3 x 45 socket head	22	J3_p2	J3_p1 / clear
split_J3_3	M3 x 45 socket head	22	J3_p2	J3_p1 / clear
collar_clamp_158_+1	M3 x 45 socket head	23	collar_upper_+y	collar_upper_-y / clear
collar_clamp_158_-1	M3 x 45 socket head	23	collar_upper_+y	collar_upper_-y / clear
collar_clamp_299_+1	M3 x 45 socket head	23	collar_fore_+y	collar_fore_-y / clear
collar_clamp_299_-1	M3 x 45 socket head	23	collar_fore_+y	collar_fore_-y / clear
pin_J2_collar_+1	M3 x 12 socket head	24	-	- / clear
pin_J2_collar_-1	M3 x 12 socket head	24	-	- / clear
pin_J2_turret_+1	M3 x 12 socket head	24	-	- / clear
pin_J2_turret_-1	M3 x 12 socket head	24	-	- / clear
pin_J3_collar_+1	M3 x 12 socket head	24	-	- / clear
pin_J3_collar_-1	M3 x 12 socket head	24	-	- / clear
pin_J3_fork_+1	M3 x 12 socket head	24	-	- / clear
pin_J3_fork_-1	M3 x 12 socket head	24	-	- / clear

5 Bus cables: one joint per cable, loops

cable	crosses	range	exit r from axis	span min..max (mm)	change	loop / cable
J1 -> J2	J1	+-90.0	40.0 mm (bound 80)	144.6 .. 179.0	34.5 mm	54 / 263 mm
J2 -> J3	J2	+-54.0	14.8 mm (bound 24)	184.1 .. 200.7	16.6 mm	37 / 267 mm
J3 -> J4	J3	+-72.0	14.8 mm (bound 28)	141.6 .. 155.1	13.5 mm	33 / 219 mm
J4 -> J5	J4	+-90.0	40.0 mm (bound 80)	106.8 .. 148.4	41.6 mm	62 / 240 mm
J5 -> J6	J5	+-46.5	14.8 mm (bound 21)	74.6 .. 85.3	10.7 mm	31 / 146 mm

controller -> J1: both in the foot, no joint crossed

J6 is the last servo on the bus: no cable leaves it

No joint turns more than 180 deg in total, so no cable can wind up; the largest path change is 42 mm.

Tie each cable down on both sides of its joint and leave the loop shown.

Move every joint through its full range by hand before powering up.

6 Service

- * To take a servo out: the order of its build step reversed (every path checked free, with our servo model and with the manufacturer's).
- * J4 servo screws: under the J4 base plate; in the finished arm the forearm spring collars cover them -- take the collars off first.
- * Base -> foot M4 bolts 2 and 3 sit under the J2 pod at J1 = 0: turn J1 to +45 / -45 deg.
- * Watch the servos' temperature over the bus (J2 works near its continuous rating).

What this manual does not tell you: tuning of the servos (PID, speed, torque limits) and the control software. The design documents: ARM450_AUDIT_REPORT.pdf (read first), ARM450_REV_I_FINAL.pdf (sections + logs), ARM450_POSES.pdf, ARM450_PARTS.pdf, LOG_SUMMARY.pdf.